

Project 1: Classification in Practice

The Big Question

How can we build and evaluate classification models to make predictions about a real-world outcome?

Choose **ONE** of the three datasets provided.

Your goal is not simply to find the model with the highest accuracy. You will investigate the data, build several classifiers, evaluate their performance, and explain what the results mean **in the context of your problem**.

Dataset Choices

Option A — Breast Cancer

Response: Malignant or benign

Scenario: You are analyzing measurements collected from breast masses. Your goal is to determine whether these measurements can be used to classify a mass as malignant or benign. Think carefully about the consequences of incorrectly classifying a malignant tumor as benign.

Option B — Heart Disease

Response: Heart disease present or absent

Scenario: You are analyzing patient characteristics and clinical measurements. Your goal is to determine whether these variables can help classify patients according to the presence or absence of heart disease. Think carefully about the consequences of failing to identify someone who actually has heart disease.

Option C — Bank Marketing

Response: Customer subscribes or does not subscribe

Scenario: A bank has conducted a marketing campaign. Your goal is to determine whether customer characteristics and campaign information can help predict whether a customer will subscribe to the product offered. Think about the costs of contacting customers who are unlikely to subscribe versus missing customers who would have subscribed.

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Part 1: Understand Your Problem

Before modeling anything, understand what you are trying to predict.

1. What does each row of your dataset represent? What is your response variable? What are its possible classes?
 2. Select the predictors you believe could be useful for classification.
 3. Examine the distribution of your response variable.
 - How many observations belong to each class?
 - Is the dataset reasonably balanced or imbalanced?
 4. Define a **false positive** and **false negative in the context of your dataset**.
 5. Which type of error would you be more concerned about in this application, or are they similarly important? Explain your reasoning.
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Part 2: Explore and Prepare the Data

1. Investigate your features and predictors.
 2. Determine whether your dataset contains missing values. If it does, explain how you handled them.
 3. Determine whether any categorical predictors need to be encoded before modeling.
 4. Divide the data, then use k-fold CV on training data.
 5. Determine whether predictors should be standardized for any of the classifiers you will use.
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Part 3: Build Your Classifiers

You will compare the four classification methods covered in DATA 201:

Logistic Regression, LDA, QDA, KNN

Then choose the one that fits. Explain why you chose your classifier.

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For Logistic Regression: Fit a logistic regression classifier using your training data. Report the model's test predictions and predicted probabilities.

For LDA: Fit an LDA classifier using the same training data and predictors. Generate predictions for the test data.

For QDA: Fit a QDA classifier using the same training data and predictors. Generate predictions for the test data.

For KNN: KNN requires you to choose **K**. Do not simply choose a value. Use **cross-validation on the training data** to compare several reasonable values of K. Create a graph showing: **K → Cross-validation performance**

Which K did you select, and why? Then fit your final KNN classifier using that value.

Part 4: Evaluate Your Models

Now compare the four models using the **test data**.

1. Produce a confusion matrix for each model. Choose a threshold and explain why you chose it.
2. Calculate the following for each model:

Accuracy, Precision, Recall/Sensitivity, Specificity, F1 score, ROC-AUC

Complete a table:

Model	Accuracy	Precision	Recall	Specificity	F1	ROC-AUC
Logistic Regression						
LDA						
QDA						
KNN						

Part 5: Reflect

Explain what you discover. Based on the consequences of false positives and false negatives that you discussed in Part 1, which metrics are particularly informative for this application? Explain.

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Part 6: ROC and AUC

Plot its ROC curve and report its ROC-AUC. Explain in your own words what ROC-AUC tells us.

Part 7: Final Analysis

Imagine that you are presenting your findings to someone who understands the problem but **has not taken DATA 201**.

- 1- Compare the performance of your four classifiers. What are the most important differences?
- 2- Identify the model you would deploy for this specific application and justify the choice using your results and the consequences of classification errors.
- 3- Explain one limitation of your analysis.

Examples could include:

- dataset size
- class imbalance
- available predictors
- missing information

4. What additional information or data would you want before using this model in the real world?

Part 8: Presentation: The Model Pitch

3-5 minute Model Pitch. Maximum 4 slides:

Slide 1- The Problem: What are you predicting, and why does the classification problem matter?

Slide 2 -The Data: Show one meaningful visualization and briefly describe the dataset.

Slide 3 - The Models: Show the four-model comparison table and ROC curve. Highlight the most important result.

Slide 4 : Your Decision Which model would you deploy and why?

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Discuss the metric(s) that drove the decision, the consequences of false positives/false negatives, and one limitation.

Post your presentation video on blackboard.

Deliverables

1. **Google Colab notebook** — all code, outputs, visualizations, and analysis.
2. **GitHub repository** – this should be new repository called “Data 201-Project1” It should include a completed notebook, the dataset, and a short README.md describing the project and main finding.
3. **4-slide / 5-minute Model Pitch** -posted on blackboard.

Grading

Project 1 is **15% of the course grade**:

Component	Points
Problem understanding & data exploration	15
Data preparation	10
Logistic, LDA, QDA & KNN implementation	20
Cross-validation & K selection	10
Model evaluation	20
ROC & threshold analysis	10
Final interpretation	10
Model Pitch	5
Total	100

Code alone does not earn full credit. You need to interpret the results.